

University of St. Gallen

School of Management, Economics, Law, Social Sciences, International
Affairs and Computer Science

Stochastic Methods in Finance

Assignment – Spring Semester 2026

Group members:

Erwan Rollin

Andrea Cimoli

Juliette Binz

Mathis Clement

Supervisor: Prof. Enrico De Giorgi

Submitted: 18 May 2026

Contents

1	Introduction	2
2	Building the Binomial Tree	2
3	Arithmetic Average at Terminal Nodes	2
4	Payoff at Terminal Nodes	2
5	Payoff at Terminal Nodes	2
6	Risk-Neutral Probabilities	2
7	Backward Induction and Option Price	2
8	Robustness Check	2
9	Normal Approximation	2
10	Conclusion	2

1 Introduction

2 Building the Binomial Tree

3 Arithmetic Average at Terminal Nodes

4 Payoff at Terminal Nodes

Table 1 reports descriptive statistics of the payoff $A_T = \max(S_n - \bar{S}_n, 0)$ computed at the terminal nodes of the binomial tree ($t = 25$).

Statistic	Value (\$)
Count	3.36×10^7
Mean	15.71
Std	24.71
Min	0.00
25%	0.00
Median	0.00
75%	24.65
Max	321.58

Table 1: Descriptive statistics of the payoff A_T at the terminal nodes ($t = 25$).

The distribution of A_T is highly right-skewed: the median is zero, meaning more than half of all terminal paths yield a worthless option, i.e. $S_n < \bar{S}_n$. Notably, the standard deviation (24.71) exceeds the mean (15.71), reflecting the high dispersion of payoffs. Most paths expire worthless while a small number of highly favourable paths drive the mean upward. The maximum payoff of 321.58 corresponds to the path where the stock price increases at every period.

5 Payoff at Terminal Nodes

6 Risk-Neutral Probabilities

7 Backward Induction and Option Price

8 Robustness Check

9 Normal Approximation

10 Conclusion